Starting Together, Diverging Later?

Gender Differences in Universal Pre-K's Long-Term Effects

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Motivation

- **Does attending public pre-K improve long-run outcomes? And for whom?**
- Existing long-run evidence largely relies on **ITT** estimates of *exposure* to UPK expansions
  - Data linking individual enrollment to long-term outcomes are rarely available
  - ITT conflates heterogeneity in *treatment effects* with heterogeneity in *take-up*

$\Rightarrow$ We know little about the long-run *treatment* effects of *attending* UPK

This Paper

- Link administrative data on age-three UPK enrollment to long-run outcomes to estimate **treatment effects** of attending pre-K
- Use a family fixed-effects design, leveraging a UPK expansion in Arab municipalities in Israel (early 2000s) that generated within-household variation in enrollment
- Test potential mechanisms for the observed sex differences in returns to UPK

Preview of findings:

1. Pre-K at age 3 (vs. 4) improves school progression; limited effects on HS achievements
2. Girls benefit substantially more than boys
3. Gender gap driven by unequal home-care environments, not developmental readiness

Contributions

I. Long-run treatment effects of UPK

- Most long-run studies estimate *access*/ITT effects (Havnes & Mogstad 2011; Baker et al. 2019; DeMalach & Schlosser 2024)
- Few estimate *treatment* effects (Gray-Lobe et al. 2022; Akee & Clark 2024; Head Start FE: Currie & Thomas 1995, Deming 2009)

II. Mechanisms behind gender differences in returns to pre-K

- Many studies find girls benefit more, but *why*? (Havnes & Mogstad 2011; Cornelissen et al. 2018; Felfe et al. 2015)
- I test two hypotheses: differential developmental readiness (Reeves 2022) vs. differences in counterfactual home-care environments (Barcellos et al. 2014; Jayachandran & Pande 2017)

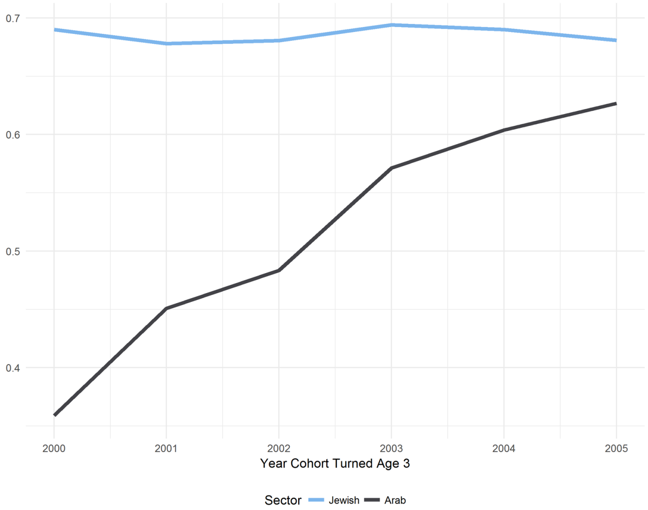
Background

Israel's UPK expansion and the Arab sector

Background: Pre-K in Israel

- Public Pre-K in Israel
  - Starts at age 3
  - Run by municipalities
  - Regulated by the government
  - Full day
- Before 1999, many Arab-majority municipalities underprovided pre-K services
- Early 2000s: a dramatic expansion (two stages: 2000/99 and 2001/02)

Pre-K Enrollment



Background: Arab Citizens of Israel

- Approximately 20% of Israel's population; 83% Muslim
- Women's labor force participation: 24% (vs. 79% for Jewish non-Ultra-Orthodox women, 2004)
- **81%** of Arab children aged 0–5 were cared for at home by a parent or unpaid relative
- $<$1% of Arab 3-year-olds attended private pre-K programs

$\Rightarrow$ Alternative to public pre-K: **home care by mothers or relatives**

Data and Sample

- Data (2000-2019):
  - Background information (parent ID, parent education, sex, DOB)
  - Enrollment in Pre-K
  - School progression (grade repetition, dropout)
  - High school achievements (*Bagrut* diploma, diploma quality)
- Sample restrictions:
  - 33 Arab-majority towns included in the second stage of the expansion
  - Birth cohorts 1996–2001 (aged 3 during school years 1999/2000–2004/5)
  - 55,264 students ($\sim$ 30% of Arab students) summary stats

Selection

	Dependent variable: Pre-K Enrollment			
	Univariate	Multivariate Models		
	(1)	(2)	(3)	(4)
Girl	-0.001	0.001	-0.001	0.004
	(0.003)	(0.003)	(0.003)	(0.004)
Father Schooling > 10	0.014^{***}	0.009^{***}	0.019^{***}	-
	(0.003)	(0.003)	(0.004)	
Mother Schooling > 10	0.118^{***}	0.068^{***}	0.029^{***}	-
	(0.003)	(0.004)	(0.008)	
Family Size	-0.033^{***}	-0.039^{***}	-0.003	-
	(0.001)	(0.003)	(0.003)	
Month of Birth	-0.007^{***}	-0.007^{***}	-0.007^{***}	-0.007^{***}
	(0.0005)	(0.0005)	(0.001)	(0.001)
Birth Order	-0.025^{***}	0.024^{***}	-0.004	0.003
	(0.001)	(0.002)	(0.003)	(0.003)
Municipality–Cohort Fixed Effects	✗	✗	✓	✓
Family Fixed Effects	✗	✗	✗	✓
Observations	54,264	54,264	54,264	40,644

within-family OLS

Empirical Strategy

- **Core idea:** compare siblings within the same family who differ in pre-K attendance because of the staggered rollout

- I estimate the following fixed effects model

$$y_i = \alpha + \beta PreK_i + \gamma X_i + \delta_{j(i)} + \theta_{k(i)t(i)} + \epsilon_i \tag{1}$$

  - $\delta_{j(i)}$ - family fixed effect $\theta_{k(i)t(i)}$ - municipality$\times$cohort fixed effect
  - X_i: sex, month of birth, birth order

- Identifying assumption: within-family variation in pre-K attendance is uncorrelated with unobservables

- **Interpretation:** effect of starting pre-K at age 3 *rather than age 4* (most non-attenders at age 3 attend at age 4)

Who Are the Families Who Switch?

- Only families with variation in pre-K (*switchers*) identify β (Miller et al. 2023)
- I define four types of families:
    1. Never Takers - send none of their children to pre-K
    2. Always Takers - send all of their children to pre-K
    3. Expansion Compliers - send the youngest child to pre-K and not the oldest child
    4. Expansion Defiers - send the oldest child to pre-K and not the youngest child

Who Are the Families Who Switch?

	Non-switchers		Switchers	
	Never Takers	Always Takers	Expansion Compliers	Expansion Defiers
	(1)	(2)	(3)	(4)
Number of Families	1,430	11,982	2,608	727
Children in Sample per Family	2.58	2.37	2.63	2.47
Father Years of Schooling	9.87	10.44	9.92	10.08
Mother Years of Schooling	9.09	10.37	9.53	9.61
Young Child Born Sep-Dec (%)	44.8%	38.8%	35.3%	50.6%
Birth Order	3.99	2.83	3.69	3.50
Family Size	4.82	3.49	4.45	4.19
Oldest Child Attended				
Pre-K at Age 3	0	1	0	1
Pre-K at Age 4	0.40	0.97	0.73	0.85

- About 80% of switchers are expansion compliers $\rightarrow$ This is not the case in non-expansion towns results
- Expansion defiers are more likely to have a younger child born Sep-Dec

Results

Pre-K improves school progression; girls benefit most

Results - Effect on School Progression

Pre-K significantly improves school progression

	First Grade on time	Grade Repeat	11^{th} Grade	**Grad HS on Time**	Grad HS
Estimate	0.019***	−0.013*	0.009	**0.024****	0.008
	(0.004)	(0.007)	(0.007)	**(0.008)**	(0.007)
Mean	0.979	0.108	0.879	0.820	0.852
Observations	40,644	40,644	40,644	40,644	40,644
Adjusted R^2	0.077	0.237	0.293	0.305	0.309

$\longrightarrow$ Pre-K increases on-time HS graduation by better progression throughout the k-12 system

Results - HS Diploma Outcomes

Limited effects on educational achievements

	HS Diploma (Bagrut)	Diploma Credits	Advanced Math	Advanced English	Advanced Hebrew
Estimate	0.003	0.442**	−0.007	0.012	0.010
	(0.009)	(0.215)	(0.007)	(0.009)	(0.008)
Mean	0.527	19.403	0.213	0.466	0.312
Observations	40,644	40,644	40,644	40,644	40,644
Adjusted R^2	0.396	0.549	0.436	0.476	0.439

$\longrightarrow$ Pre-K improves school progression but not high school achievements — can rule out effect sizes of previous studies

Heterogeneity - Sex

Girls benefit substantially more than boys across all progression outcomes

	First Grade On Time	Grade Repeat	11^{th} Grade	Grad HS on Time	Grad HS
Boys	0.016^{***}	-0.007	-0.017^{*}	0.006	-0.014
	(0.005)	(0.009)	(0.009)	(0.011)	(0.010)
Girls	**0.022*****	**−0.018****	**0.036*****	**0.042*****	**0.030*****
	(0.004)	**(0.008)**	**(0.008)**	**(0.009)**	**(0.009)**
p-value for equal effects	0.240	0.240	**0.000**	**0.002**	**0.000**
Joint p-value across outcomes			**6.03e−10**		

$\longrightarrow$ Despite identical participation rates, girls' effects are 3–7$\times$ larger; boys show no gains beyond 1st grade entry

Who Benefits More from UPK?

Sex

Girls benefit substantially more from UPK

Joint p-value $< 10^{-9}$

Relative Age

results

Younger children gain more than older ones

Reverse selection on gains

Mother's Education

results

Children of more-educated mothers gain more

In low-education families, **only girls** benefit

Family Size

results

Larger families show modestly larger gains

Mechanisms

Testing developmental readiness vs. home-care environment

What Can Explain Sex Heterogeneity?

H1: Differential developmental readiness

Girls are more developmentally ready to benefit from pre-K

$\rightarrow$ *Prediction: older boys should gain more from UPK than younger boys*

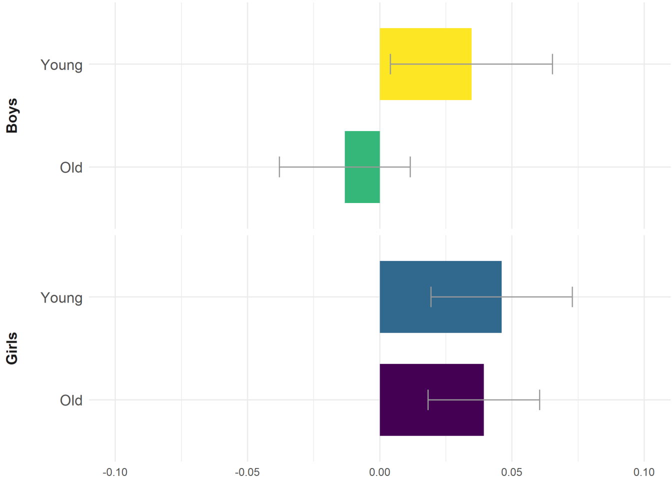
H2: Differences in counterfactual home-care environment

Gender norms and son preference shape children's daily experiences

Son preference may skew within-household resource allocation toward boys

Heterogeneity by Sex and Relative Age

The Effect of Attending UPK on Graduating HS on time



What Can Explain Sex Heterogeneity?

✗ **H1: Differential developmental readiness** — *contradicted by evidence*

H2: Differences in counterfactual home-care environment

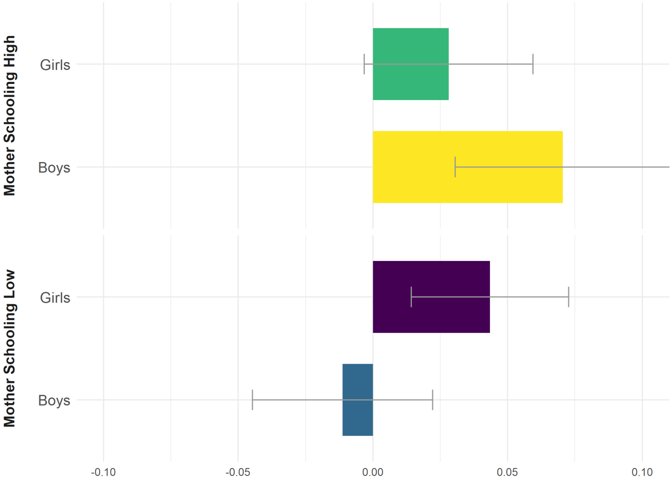
Gender norms and son preference shape children's daily experiences and parental investments

→ *Prediction: sex heterogeneity should be stronger in more traditional families (proxied by maternal education)*

Son preference may skew within-household resource allocation toward boys

Heterogeneity by Sex and Maternal Schooling

The Effect of Attending UPK on Graduating HS on time



What Can Explain Sex Heterogeneity?

✗ **H1: Differential developmental readiness** — *contradicted by evidence*

H2: Differences in counterfactual home-care environment ✓

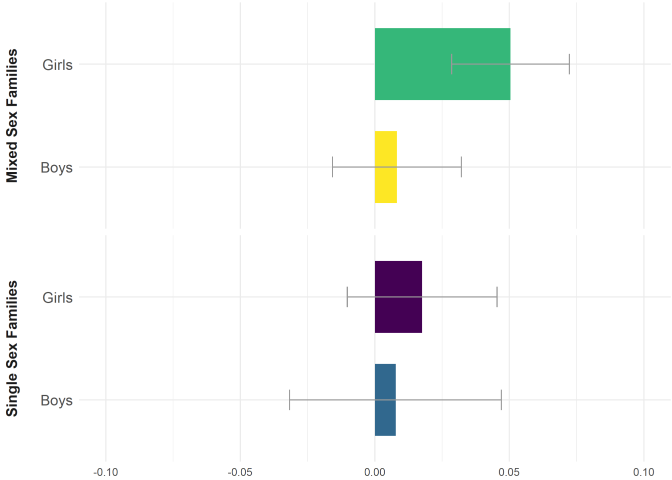
Gender norms and son preference shape children's daily experiences — *confirmed*

Son preference may skew within-household resource allocation toward boys

$\rightarrow$ *Prediction: sex heterogeneity should be stronger in mixed-sex families than in same-sex families*

Heterogeneity by Sex and Household Composition

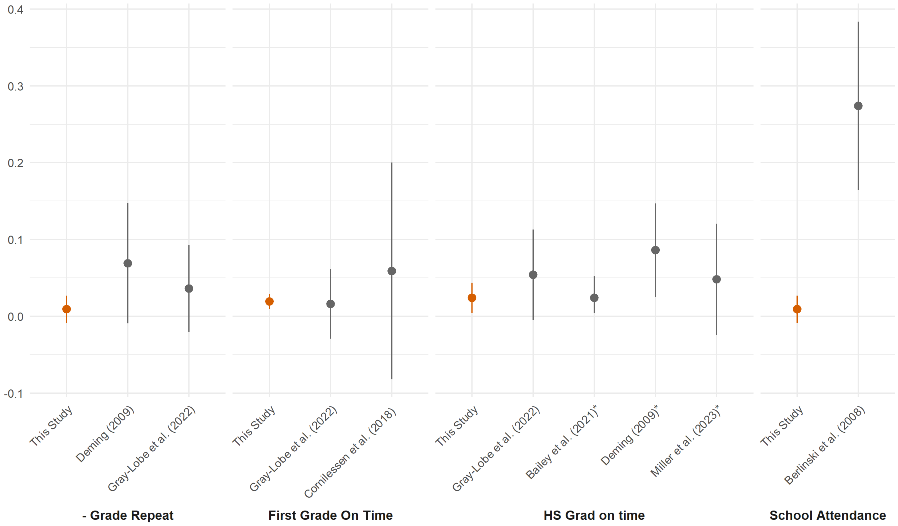
The Effect of Attending UPK on Graduating HS on time



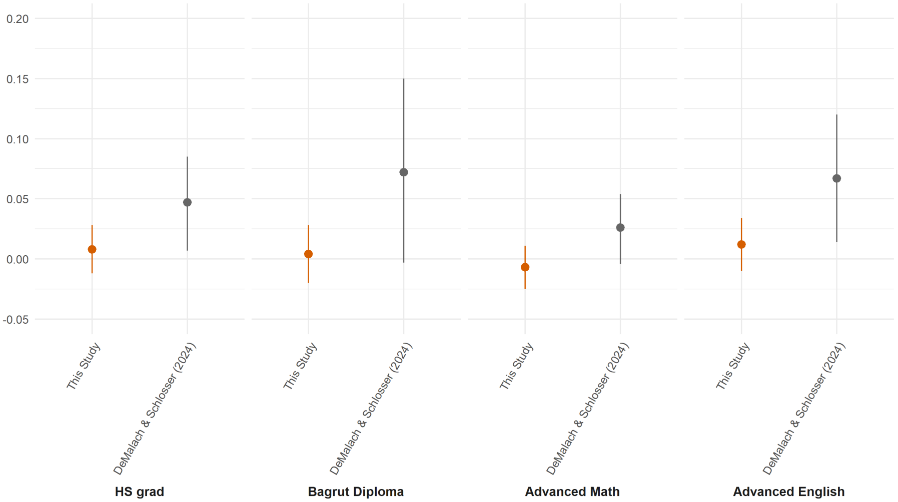
Additional Results and Robustness

- Additional Results
  - No heterogeneity in educational achievement outcomes results
  - No evidence for spillover to siblings results
- Robustness
  - Excluding expansion defiers robustness
  - Specifications (no controls for month of birth; common cohort FEs) robustness
  - Sample of towns all

Effect Size - Comparison to Literature



Comparison to DeMalach & Schlosser (2024)



Conclusions

- Long-run *treatment effect* estimates of UPK using admin data and family FE
- Pre-K at age 3 (vs. 4) improves school progression; limited effects on HS achievement
- **Girls benefit more than boys** across all progression outcomes; gender gap is driven by unequal home-care environments, not developmental readiness
- Effects are smaller than previous ITT studies — consistent with
  (a) this being a marginal year (age 3 vs. 4), and (b) ITT capturing spillovers on non-participants
- MVPF well below 1 from progression gains alone (Hendren & Sprung-Keyser 2020); full social return likely higher

> $\Rightarrow$ **UPK as an equalizer:** pre-K compensates for unequal home environments, but in doing so widens the gender gap in educational outcomes

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Expansion Compliers by MOB

Expansion Towns

Younger Sibling	Older Sibling Jan-Aug	Older Sibling Sep-Dec
Jan-Aug	80.0%	87.4%
Sep-Dec	68.5%	76.9%

Non-Expansion Towns

Younger Sibling	Older Sibling Jan-Aug	Older Sibling Sep-Dec
Jan-Aug	50.2%	56.6%
Sep-Dec	44.3%	54.4%

back

Heterogeneity - Relative Age

	First Grade On Time	Grade Repeat	11^{th} Grade	Grad HS on Time	Grad HS
Young	0.026^{***}	-0.018^{*}	0.016^{*}	0.040^{***}	0.011
	(0.006)	(0.009)	(0.009)	(0.011)	(0.010)
Old	0.014^{***}	-0.009	0.005	0.013	0.005
	(0.004)	(0.009)	(0.008)	(0.011)	(0.009)
p-value for equal effects	0.043	0.369	0.263	0.026	0.628
Joint *p*-value across outcomes			0.0448		

more back

Heterogeneity - Mother Education

	First Grade On Time	Grade Repeat	11^{th} Grade	Grad HS on Time	Grad HS
High	0.039^{***}	-0.021^{*}	0.016	0.050^{***}	0.024^{**}
	(0.007)	(0.012)	(0.011)	(0.014)	(0.012)
Low	0.014^{**}	-0.017	0.015	0.016	0.001
	(0.005)	(0.011)	(0.011)	(0.012)	(0.011)
p-value for equal effects	0.003	0.810	0.978	0.057	0.131
Joint p-value across outcomes			0.0121		

more back

Heterogeneity - Family Size

	First Grade On Time	Grade Repeat	11^{th} Grade	Grad HS on Time	Grad HS
Small	0.016***	−0.008	−0.011	0.020*	0.003
	(0.005)	(0.009)	(0.009)	(0.011)	(0.010)
Large	0.021***	−0.016*	0.023***	0.027***	0.011
	(0.005)	(0.009)	(0.009)	(0.009)	(0.009)
p-value for equal effects	0.401	0.472	0.003	0.595	0.503
Joint *p*-value across outcomes			0.0516		

back

Heterogeneity in HS Diploma Outcomes

	HS Diploma (Bagrut)	Diploma Credits	Advanced Math	Advanced English	Advanced Hebrew
Boys	0.002	0.619*	−0.001	0.013	0.003
	(0.015)	(0.340)	(0.012)	(0.014)	(0.014)
Girls	0.006	0.357	−0.012	0.011	0.019
	(0.015)	(0.353)	(0.010)	(0.013)	(0.013)

back

Summary Stats

	Entire Sample (1)	Family FE Sample (2)
Attended Pre-K	0.815	0.806
	(0.388)	(0.395)
Girl	0.489	0.499
	(0.500)	(0.500)
Father Schooling	10.296	10.241
	(2.778)	(2.723)
Mother Schooling	10.092	10.008
	(2.611)	(2.555)
Family Size	3.784	3.915
	(1.954)	(1.973)
Month of Birth	6.621	6.626
	(3.418)	(3.423)
Birth Order	3.144	3.161
	(2.001)	(1.995)
First Grade on Time	0.980	0.979
	(0.141)	(0.143)
Graduated High School on Time	0.826	0.820
	(0.379)	(0.385)
Graduated High School	0.858	0.852
	(0.349)	(0.355)
Attended 11th Grade	0.884	0.879
	(0.321)	(0.326)
Ever Repeated a Grade	0.106	0.108
	(0.308)	(0.310)
Diploma (Bagrut)	0.540	0.527
	(0.498)	(0.499)
Diploma Credits	19.745	19.403
	(12.840)	(12.872)
Advanced Math	0.223	0.213
	(0.417)	(0.409)
Advanced English	0.483	0.466
	(0.500)	(0.499)
Advanced Hebrew	0.316	0.312
	(0.465)	(0.463)
Number of Students	54,264	40,644
Number of Families	30,466	16,747

back

OLS Estimates

	Dependent Variable: High School Diploma (*Bagrut*)				
	(1)	(2)	(3)	(4)	(5)
PreK-3	0.124***	0.089***	0.056***	0.050***	0.003
	(0.006)	(0.007)	(0.006)	(0.007)	(0.009)
Background Controls	✗	✓	✓	✓	✓
Municipality-Cohort FE	✗	✗	✓	✓	✓
Siblings Sample	✗	✗	✗	✓	✓
Family FE	✗	✗	✗	✗	✓
Observations	54,264	54,264	54,264	40,644	40,644
Adjusted R^2	0.009	0.058	0.195	0.194	0.396

back

Siblings Spillovers

	Panel A: Heterogeneity by Sibling Spacing	
	(1)	(2)
PreK	0.023	0.024
	(0.008)	(0.015)
PreK $\times$ Age Gap	0.007	–
	(0.009)	
PreK $\times 1\{\text{Age Gap} > \text{Median}\}$	–	-0.005
		(0.019)
Number of students	34,182	34,182
Number of families	16,747	16,747

	Panel B: Difference-in-Differences	
	Main Sample (1)	All Arab-majority Towns (2)
Child 2 $\times$ Treated	0.001	-0.007
	(0.020)	(0.008)
Number of students	8,966	29,916
Number of families	4,483	14,966

back

Robustness

Robustness Test	HS Grad On Time	HS Diploma (Bagrut)
No Expansion Defiers	0.020**	−0.005
	(0.009)	(0.011)
Number of Observations	38,846	
Common Cohort Fixed Effect	0.025***	−0.005
	(0.007)	(0.009)
Number of Observations	40,644	
No Control for Month of Birth	0.028***	0.006
	(0.008)	(0.009)
Number of Observations	40,644	
All Arab-majority Municipalities	0.015***	0.001
	(0.004)	(0.005)
Number of Observations	133,751	
Only Youngest and Oldest Child	0.023**	0.005
	(0.009)	(0.011)
Number of Observations	34,182	

back

All Arabs Towns

	Panel A: School-Progression Outcomes				
	First Grade on time (1)	Grade Repeat (2)	11th Grade (3)	Grad HS on Time (4)	Grad HS (5)
Estimate	0.006***	−0.002	0.005	0.015***	0.009**
	(0.002)	(0.004)	(0.004)	(0.004)	(0.004)
Mean	0.978	0.107	0.845	0.782	0.813
Observations	133,751	133,751	133,751	133,751	133,751
Number of Families	55,138	55,138	55,138	55,138	55,138
Adjusted R^2	0.109	0.217	0.315	0.334	0.336
	Panel B: HS Diploma Outcomes				
	HS Diploma (Bagrut) (1)	Diploma Credits (2)	Advanced Math (3)	Advanced English (4)	Advanced Hebrew (5)
Estimate	−0.001	0.179	−0.003	0.004	0.001
	(0.005)	(0.111)	(0.004)	(0.004)	(0.004)
Mean	0.437	16.214	0.172	0.387	0.250
Observations	133,751	133,751	133,751	133,751	133,751
Number of Families	55,138	55,138	55,138	55,138	55,138
Adjusted R^2	0.466	0.614	0.442	0.514	0.447

back

Within-Family Selection: Birth Weight & Maternal Employment

	Panel A: Child Birth Weight		
	Mean [SD] (1)	No Family FE (2)	Family FE (3)
Birth weight (kg)	3.23 [0.52]	0.012*** (0.002)	0.004 (0.003)
Family Fixed Effects		✗	✓
Number of children		144,348	
Number of families		61,913	

	Panel B: Maternal Labor Supply		
	Mean [SD] (1)	No Family FE (2)	Family FE (3)
Mother worked	0.27 [0.44]	0.060*** (0.003)	0.003 (0.004)
Months worked	2.39 [4.41]	0.0054*** (0.0003)	0.0004 (0.0004)
Mother's earnings (in NIS 1,000)	10.11 [21.45]	0.0008*** (0.0001)	−0.00007 (0.0001)
Family Fixed Effects		✗	✓
Number of children		151,773	
Number of families		66,441	

⟶ Cross-sectional correlations vanish with family FE — within-family enrollment is not driven by child endowments or household circumstances